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Literature Review

Digital Art Gallery Platform (A Platform to Showcase, Buy, and Sell Local Artworks)

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Contents

1. INTRODUCTION	3
2. Digital Platforms for Art Accessibility and Market Expansion	3
3. Key Challenges in Traditional Art Markets and Opportunities for Digital Transformation	4
4. Existing Implementations of Online Art Gallery Systems	5
5. Immersive 3D Virtual Gallery Experiences and User Engagement	6
6. Recommendation Systems for Artwork Discovery and Personalisation.....	6
7. Market Analysis of Existing Art Platforms.....	7
International Platforms.....	7
Local Scenario (Nepal).....	7
Competitive Advantage:	7
8. Advanced Inventory Management Strategies in Digital Art Platforms.....	8
9. Digital and Predictive Approaches to Art Gallery Inventory Management	8
10. Similar Systems	9
10.1 Saatchi Art.....	9
10.2 Etsy (Handmade & Art Section)	9
10.3 ArtPal.....	10
11. Conclusion.....	11
Key findings from the research include:	11
The Digital Art Gallery Platform uniquely addresses these gaps by:	11
12. References	12

1. INTRODUCTION

Digital transformation has reshaped the way creative industries operate worldwide, yet many local artists in developing regions like Nepal still lack access to professional platforms that support global visibility, secure transactions, and sustainable income generation. Physical galleries are confined to major urban centers and are expensive to access as an exhibition space, making them unavailable for emerging and rural artists. Current reliance on informal social media channels restricts credibility and does not provide any structured commerce mechanism; it also creates scant opportunities for audience engagement and data-driven insights. Consequently, artist exposure remains at a minimum, economic opportunities are constrained, and much of Nepal's cultural heritage remains underrepresented in digital spaces. Mindful of these challenges, the proposed Digital Art Gallery Platform aims to create an accessible, professionally structured, and technologically enriched web-based environment where artists can showcase their work, reach global audiences, utilize immersive 3D virtual galleries, and receive AI-powered recommendations for better visibility—all this while being able to securely process payments via locally trusted gateways like Khalti. Therefore, the project aims to bridge the gap between traditional artistic expression and modern digital commerce, increasing the level of accessibility, engagement, and economic empowerment for Nepali artists.

2. Digital Platforms for Art Accessibility and Market Expansion

Recent developments in digital commerce and online cultural ecosystems indicate a major shift toward web-based platforms as an effective solution for improving accessibility, visibility, and economic opportunities for artists, particularly those in developing regions. Studies on digital marketplaces highlight that virtual platforms remove geographic barriers, enabling artists from remote or underserved areas to reach international audiences without the high operational expenses associated with traditional galleries. This shift is especially significant in countries like Nepal, where physical art exhibitions are concentrated in a few metropolitan areas, restricting access for rural or emerging artists.

Furthermore, studies on how consumers behave in digital settings show that professionally designed online platforms with well-organized artwork practices lack formal customer support systems, pricing transparency, and verification mechanisms. As a result, digital platforms lower the risk of fraud while boosting market credibility, making the environment more dependable for both buyers and artists. Additionally, because they increase user engagement, immersive virtual experiences like 3D galleries have become more well-known. Compared to static 2D

listings, users spend a lot more time exploring artworks in interactive digital spaces, according to recent UX studies.. This increase in engagement is strongly correlated with higher emotional connection, improved aesthetic appreciation, and increased purchase intent. Technologies like WebGL and Three.js further enable realistic gallery environments where viewers can move through virtual spaces, zoom into artworks, and interact with pieces in ways that mimic real-life exhibitions.

Moreover, incorporating AI-powered recommendation systems is essential to improving the discoverability of artwork. By offering tailored recommendations that are in line with personal preferences, recommendation engines—whether they are based on collaborative filtering, content attributes, or hybrid models—have been demonstrated to boost user satisfaction. This feature is especially helpful for small or independent artists because AI exposure guarantees equal visibility across a variety of artistic styles and backgrounds and lessens reliance on popularity-based rankings. When taken as a whole, digital art platforms that incorporate secure purchasing methods, intelligent recommendations, and immersive design present a revolutionary way to modernize the art market. These platforms not only increase revenue potential and audience reach, but they also promote international appreciation of regional artistic talent and wider cultural exchange.

3. Key Challenges in Traditional Art Markets and Opportunities for Digital Transformation

Research into creative industry dynamics in developing countries repeatedly points out a number of systemic challenges that exist within the traditional art market. Physical galleries often have limited capacities, high costs for exhibiting works, and strong urban concentration, making them inaccessible for the majority of emergent creators. Most artists cannot afford the financial outlay associated with traditional exhibitions, such as rental costs, transportation, framing, installation, and promotional activities. This provides a significant cause of inequality, whereby only established artists who have financial support can regularly show their work.

Moreover, traditional gallery systems are deeply dependent on a regime of programmed exhibitions and seasonal events, preventing continuous visibility for artists. Consequences include the shortening of the life cycle of an artwork's exposure to only a few weeks, after which it may not be seen by prospective buyers. This further leads to lower opportunities for sale, reduced income, and instability in the long-term career building of artists.

Meanwhile, the majority of budding artists take the help of informal digital channels like Facebook, Instagram, or WhatsApp in order to popularize and sell their work. However, these channels lack the basic facilities that include secure payment gateways, professional

presentation layouts, transactional transparency, customer analytics, and fraud prevention mechanisms. As far as buyers are concerned, there are problems like unverified sellers, inconsistent pricing, poor communication, and no purchase protection. Aggregately, these issues weaken trust and make a sale less likely to take place.

Lack of structured digital documentation also presents a long-term threat in the area of cultural preservation. Most of the artworks and various styles, as well as many traditional skills exercised by the local artists, are either not recorded or badly cataloged. Without structured digital systems, such cultural assets face the threat of underrepresentation within global cultural discourse.

These limitations can be overcome as digital platforms provide a centralized, organized, and technologically supported environment for the art market. Digital systems create a fairer and more effective marketplace through features like artist portfolios, searchable artwork databases, secure local payment integrations such as Khalti, and automated sales management.

Additionally, AI recommendations provide more exposure to less well-known artists by helping buyers discover artworks that fit their preferences, viewing behavior, or similar user profiles. Provided with 3D virtual galleries, these tools can introduce immersive, culturally rich experiences on par with a physical exhibition, but without the limitations imposed by cost.

The intersection of these technologies enacts changes on the traditional art market, making it more inclusive, transparent, and scalable. For regions like Nepal, where cultural identity and artistic heritage are highly valued but still underrepresented in digital spheres, such transformation means not just expanded economic opportunities but also an integral part of cultural preservation and global representation.

4. Existing Implementations of Online Art Gallery Systems

Munikumari, T., & Sai Kiran, P. (2024) present a complete online art gallery system with detailed database schema design, artist and admin modules, artwork uploading mechanisms, and portfolio management features. The system emphasizes the structured categorization of artworks by medium, style, and price, along with user authentication and role-based access. The work presented serves as a foundational reference regarding core entity relationships - that is, users and artworks transactions-and it informed the MySQL schema design in this current project, especially the artworks, users, orders, and views tables. While their work focused on basic 2D presentation, it did note that extended features like immersive visualizations and intelligent discovery were needed, which the present platform follows up with via Three.js 3D Galleries and hybrid recommendation engines. This work of Betrand, C. U., Ekwealor, O. U., & Onyema, C. J. (2022) presented an Online Art Gallery Exhibition and Auction System for Nigerian

indigenous artworks using React.js, Node.js, and MongoDB. The system integrated cultural authenticity verification, secure authentication, and auction capabilities, maintaining focus on the preservation of heritage related to non-Western art. Their architecture has directly influenced the role-based JWT authentication, file upload security, and cultural context considerations in the present project, specifically the focus on Nepali artistic heritage and how artist verification is managed through workflows. This study proved the technical feasibility of full-stack art platforms in developing regions and underlined the importance of localized features, which provided a reason for integrating Khalti as the payment gateway.

5. Immersive 3D Virtual Gallery Experiences and User Engagement

Rodriguez, R. M. (2024) used the Online Gallery Aesthetics Rating (OGAR) framework to examine the psychological effects of virtual art gallery experiences. In comparison to conventional 2D image grids, the study found that 3D immersive galleries enhanced emotional engagement by 34%, prolonged average session duration by 45%, and increased purchase intent by 28%. In navigable 3D spaces, participants reported feeling more "present" and connected to the artwork. In order to replicate the demonstrated engagement benefits while preserving accessibility on low-end devices in developing regions like Nepal, these findings offer compelling empirical support for the implementation of Three.js/WebGL-based virtual galleries in the current platform, including interactive navigation (WASD/touch), clickable framed artworks, dynamic lighting, and performance optimization techniques (LOD, lazy loading, mobile texture reduction).

6. Recommendation Systems for Artwork Discovery and Personalisation

Hybrid recommendation techniques have been shown in numerous studies to be very successful at finding artwork. Comparing content-based and collaborative filtering techniques, research published in the International Advanced Research Journal in Science, Engineering, and Technology (IARJSET, 2025) found that a hybrid model that addresses cold-start and sparsity problems typical of small-to-medium art platforms achieved 83% precision and 81% recall. In addition to content-based feature extraction (HSV color histograms, style classification, medium/genre tagging), complementary works by DataCamp (2025) and Christoph Trattner offered useful implementations of user-based and item-based collaborative filtering. 15–25% accuracy gains over single-method systems were confirmed by a comprehensive literature review of more than 50 hybrid recommender systems (arXiv, 2019). These results directly

influenced the current project's streamlined but efficient hybrid recommendation engine, which combines Trending/popular queries,

- (i) Category-based matching,
- (ii) User-similarity collaborative filtering with lightweight Python logic and effective SQL that provides tailored recommendations appropriate for the Final Year Project scope while being scalable for upcoming deep-learning improvements.

7. Market Analysis of Existing Art Platforms

International Platforms			
Platform	Strengths	Weaknesses	Relevance
Saatchi Art	Global reach, professional curation, print-on-demand	35–50% commission, inaccessible to emerging artists	Shows demand for premium presentation but highlights cost barriers
Etsy	Low entry fee (6.5%), large buyer base, easy setup	Not art-specific, no 3D, minimal AI, generic tools	Proves low-cost model works but lacks immersion
ArtPal	Zero commission, social sharing, print integration	No 3D, basic search, limited analytics	Validates artist-first pricing; opportunity to add advanced features

Local Scenario (Nepal)

- No dedicated online art platform exists
- Artists rely on Facebook, Instagram, WhatsApp → informal, unprofessional
- Physical galleries: NPR 50,000–200,000 per exhibition → excludes rural/emerging artists
- Market size: USD 2–5 million annually (informal)
- Opportunity: 70%+ mobile internet penetration, growing diaspora/tourist buyers

Competitive Advantage:

- Khalti integration (local trust)
- Zero/low commission

- 3D immersion + AI
- Nepali language & cultural focus

8. Advanced Inventory Management Strategies in Digital Art Platforms

Research into multi-artwork portfolio and inventory optimization describes how dynamic stock visibility and intelligent stock allocation are needed to manage the growing number of digital and physical artworks found in online galleries. The research also deals with the challenge of substituting unavailable art works to allow the buyers to discover alternative artworks if their favorite is sold or under commission. By using a collaborative filtering-based model, the researcher proposed an almost optimal recommendation algorithm that, when compared with basic category matching, increases the relevance of discovery and engagement. The approach identifies clusters of substitutable artworks (by style, medium, or theme) without decreasing sales potential. In addition, a multi-step user journey model covers the entire buyer decision process, which includes search, 3D gallery exploration, adding to a wishlist, and finally purchasing; the approach leads to more accurate forecasts of engagement. Computational tests using both synthetic and real user interaction data show that this hybrid technique outperforms traditional grid-based browsing on both engagement duration and conversion rates.

9. Digital and Predictive Approaches to Art Gallery Inventory Management

Online platforms can make data-driven decisions about inventory exposure, artist onboarding, and artwork promotion in real time by integrating robust recommendation and visualization algorithms into a web-based digital art gallery system. This increases the discoverability of artwork, reduces lost sales opportunities, maximizes platform revenue, and raises artist compensation and cultural awareness. One important realization is that in order for art platforms to guarantee that valuable artworks are seen by the appropriate audience at the appropriate time, efficient inventory management is essential. The study highlights the application of contemporary techniques like 3D virtual gallery analytics, AI-powered recommendations, and real-time view tracking. These strategies improve overall operational efficiency, lessen the underexposure of up-and-coming artists, and help platforms promote artwork more precisely. This underscores the importance of implementing cloud-based, AI-enabled gallery systems to respond dynamically to user behavior and support sustainable digital art ecosystems in developing regions like Nepal.

(Adapted from Raj, 2025-06-13)

10. Similar Systems

10.1 Saatchi Art

Saatchi Art stands as the world's largest and most prestigious online art marketplace, hosting over 1.5 million original artworks from more than 100,000 artists across 100+ countries. It offers professional curation, global shipping logistics, and a highly polished user experience with advanced filtering by medium, style, price, and subject. Artists benefit from dedicated portfolio pages, sales analytics, and direct buyer communication. The platform's success demonstrates the commercial viability of online art commerce and the demand for professional digital presentation.

Like Saatchi Art, the Digital Art Gallery Platform enables artists to create structured portfolios, set custom prices, and sell directly to buyers. However, Saatchi imposes 35–50% commission rates, making it inaccessible to emerging and rural Nepali artists who cannot absorb such cuts. The proposed system eliminates high commissions entirely, operating on a low or zero-fee model to maximize artist earnings. Furthermore, it introduces Nepal-specific enhancements absent in Saatchi:

- 3D virtual gallery walkthroughs using Three.js
- AI-powered personalized recommendations via Python/Scikit-learn
- Khalti payment integration for seamless NPR transactions
- Nepali language support and cultural heritage tagging

These features make the platform far more inclusive and culturally relevant for underserved creators in developing regions.

10.2 Etsy (Handmade & Art Section)

Etsy is a global e-commerce leader in handmade, vintage, and creative goods, with a dedicated Art & Collectibles category supporting over 2 million sellers. It excels in low barriers to entry (6.5% transaction fee + \$0.20 listing fee), intuitive seller dashboards, and strong SEO-driven discoverability. Artists can upload high-resolution images, write detailed descriptions, and manage inventory with ease. The platform's first mobile design and social sharing tools drive high engagement.

The Digital Art Gallery Platform adopts Etsy's artist-friendly onboarding and pricing control but specializes exclusively in fine art, unlike Etsy's general-purpose craft focus. Etsy offers no 3D

immersion, no AI recommendations, and no art-specific analytics critical gaps for serious art commerce. The proposed system fills these with:

- Immersive 3D virtual galleries replacing flat image grids
- Hybrid AI recommendation engine (collaborative + content-based)
- Localized payment processing via Khalti
- Cultural preservation metadata (e.g., Thangka, Mithila, Newa art styles)

Thus, while Etsy proves low-cost models work, the current platform delivers a dedicated, high-engagement art ecosystem tailored for Nepal's artistic heritage.

10.3 ArtPal

ArtPal positions itself as an artist-first marketplace with zero commission on original sales and optional print-on-demand services through integrated partners. It supports direct messaging, social media sharing, and basic analytics (views, favorites). The platform's clean interface and no-upfront cost model have attracted thousands of independent artists seeking financial control.

The Digital Art Gallery Platform builds directly on ArtPal's zero-commission philosophy but advances significantly in technical sophistication and cultural relevance. ArtPal relies on 2D image grids and keyword search, offering no immersive experience or intelligent discovery. The proposed system introduces:

- Fully navigable 3D exhibition halls with dynamic lighting and spatial audio
- Real-time AI recommendations based on user behavior and artwork similarity
- Admin-curated cultural collections (e.g., "Women in Nepali Art" "Post-Earthquake Resilience")
- Secure, localized transactions via Khalti's verified API

By combining financial empowerment with cutting-edge visualization and personalization, the platform creates a superior alternative for Nepali artists preserving and monetizing their cultural identity.

11. Conclusion

A web-based digital art platform that incorporates 3D virtual galleries, useful AI recommendations, safe payment processing, and localized design can significantly improve accessibility, engagement, and economic opportunities for local artists in developing regions like Nepal, according to a thorough review of academic literature, technical documentation, and market analysis.

Key findings from the research include:

- Rodriguez (2024) empirically proves that 3D immersion increases emotional connection and purchase intent by over 25%
- Hybrid recommendation systems (IARJSET, 2025; DataCamp, 2025) achieve 80%+ accuracy in sparse datasets, ideal for emerging art markets
- Full-stack security best practices (OWASP, Khalti docs) ensure trust and compliance in financial transactions
- International platforms (Saatchi, Etsy, ArtPal) validate online art commerce but expose critical gaps: high fees, no immersion, no cultural focus, and no Nepal-specific infrastructure

The Digital Art Gallery Platform uniquely addresses these gaps by:

- Replacing NPR 50,000–200,000 physical exhibition costs with free, global 3D virtual galleries
- Eliminating commission barriers to ensure artists retain 95–100% of earnings
- Integrating Khalti for trusted, instant NPR payments
- Preserving Nepali heritage through structured cultural metadata and artist verification

Built on proven technologies (React.js, Node.js, MySQL, Three.js, Python) and rigorous academic foundations, this platform represents a socially impactful, technically feasible, and culturally sustainable solution empowering Nepal's artists to thrive in the global digital economy while honoring their rich artistic traditions.

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